ELEC3104: Digital Signal Processing Lab

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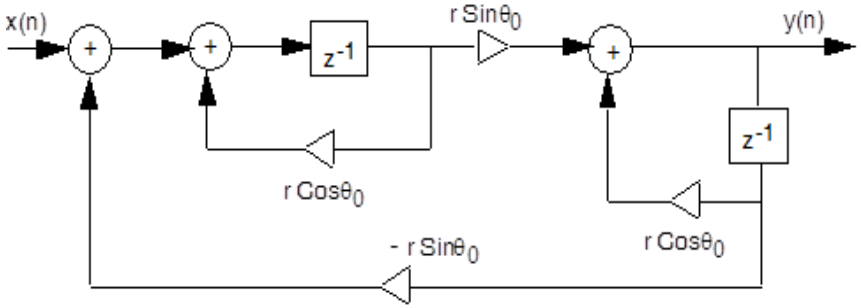
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Term 3, 2021

Topic 4 – Oscillator Implementation

Exercise 4.1: A system has the block diagram as shown below



a. Show that the transfer function of the system is given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{r\ sin\theta_0\ z^{-1}}{1-2\ r\ cos\theta_0\ z^{-1}+r^2z^{-2}}$$

b. Let $\theta_0 = \pi/4$ and find the difference equation from the above transfer function

c. When $\theta_0 = \pi/4$, r=1 and x(n) = δ(n), use **MATLAB** and choose the appropriate initial conditions in order to sustain oscillation:

   i. Write code to implement the **difference equation** of system H (z) to generate the signal y[n] when x(n) = δ(n). (NOT using the 'impz' command). Plot the first 50 samples of y[n].

   ii. Assume that $y[n]$= sin(nθ_o), verify the value of θ_o based on the plot of y[n].

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MATLAB code solution for part (c)

```
n=0:49;
x = [1, zeros(1, 49)]; % x[n] = delta(n)
th = pi/4; % th stands for theta
r = 1;
% first output sample is y(1)
% initial conditions y(0) =0;  y(-1)=0
y(1)=0; % y(1) in MATLAB correspond to y(0) in equation, y(0) = sin(0*th)=0
y(2)=r*sin(th)*x(1); %initial conditions y(0)=y(1)=0
for i=3:50
    y(i)=r*sin(th)*x(i-1) + 2*r*cos(th)*y(i-1)-r^2*y(i-2);
end
stem(n,y);
xlabel('Sample index (n)'); ylabel('y[n]');
title('Oscillator output')

%%It can be checked from the plot that each
%%period of y[n] has N=8 samples
%%As y[n]=sin(n*pi/4); N*pi/4 = 2*pi ->N=8
```

